

TrafficSim System

PeMS Data

Mainline · Ramp sensors
15-min intervals (CSV)

NetworkConfig

Segments · Lanes · GPS
Sensors · Ramp topology

DynamicsConfig

IDM | Gipps | Krauss
CF parameters · vmax

TrafficModelBuilder

VSources · Ramps · Route · Junctions · Animation coords

ScalaTion Process Engine

Arrival & Network

Multi-VSource (×4 lanes + ramps)
Poisson | Erlang2S (per-lane μ)
Route → Pathway (×N lanes)
Junction sensors · Ramps · Sink

Vehicle Physics

Car-Following Models
IDM · Gipps · Krauss

Numerical Integrators
Ballistic · Euler · Heun · RK2–RK4
Butcher · DOPRI5

Model Director / Scheduler · Discrete + Continuous Time

SimulationReport

NRMSE · RMSE · sMAPE vs Observed PeMS (lane-level · aggregate)